

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1572

Myrtle 138 kV
October 15, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1572. The Interconnection Customer proposes a 72.2 MW battery energy storage facility interconnecting at the Myrtle 138 kV substation in the Virginia area. The request was received on September 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 567 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,249,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CRITTENDEN / SMITHFIELD 138 KV
Contingency: YARNEN / MIRA VISTA 345 KV
Loading: 132.3% (Rating: 245 MVA)
DFAX: 6.7%
Pre-project loading: 130.3%
2. Facility: TOANO SUBSTATION 138 kV/1 kV Autotransformer
Contingency: YAREN SUBSTATION / MIRA VISTA 345 KV
Loading: 125.3% (Rating: 579 MVA)
DFAX: 7.9%
Pre-project loading: 124.4%
3. Facility: MEYER RD / SORENSEN 138 KV L
Contingency: BOBVILLE / AVRA VLY 138 KV
Loading: 123.2% (Rating: 221 MVA)
DFAX: 12.9%
Pre-project loading: 119.0%
4. Facility: WERNER W / YARNEN SUBSTATION 138 KV
Contingency: MCCARTNEY / YARNEN 345 KV
Loading: 118.2% (Rating: 339 MVA)
DFAX: 18.2%
Pre-project loading: 114.3%
5. Facility: SMITHFIELD / HANFOD 138 KV L
Contingency: MIRA VISTA / YARNEN 345 KV line
Loading: 115.2% (Rating: 256 MVA)
DFAX: 6.7%
Pre-project loading: 113.3%

6. Facility: SUFFOLK / MYRTLE 138 KV line
Contingency: MCCARTNEY / YARNEN 345 KV line
Loading: 113.6% (Rating: 213 MVA)
DFAX: 29.9%
Pre-project loading: 103.4%
7. Facility: MYRTLE / SANGER 138 KV
Contingency: MCCARTNEY / YARNEN 345 KV
Loading: 112.7% (Rating: 275 MVA)
DFAX: 70.1%
Pre-project loading: 94.3%
8. Facility: YARNEN 345 kV/138 kV Autotransformer
Contingency: MCCARTNEY SUBSTATION / YARNEN 345 KV
Loading: 110.1% (Rating: 381 MVA)
DFAX: 18.2%
Pre-project loading: 106.7%
9. Facility: BOBVILLE / FOOTHILLS 138 KV
Contingency: E LONGMEADOW / SANGER 138 KV
Loading: 108.8% (Rating: 204 MVA)
DFAX: 20.6%
Pre-project loading: 101.5%
10. Facility: CARROLL / CRITTENDEN SUBSTATION 138 KV Line
Contingency: YARNEN / MIRA VISTA 345 KV line
Loading: 108.3% (Rating: 375 MVA)
DFAX: 5.9%
Pre-project loading: 107.1%
11. Facility: CRITTENDEN / KRAFT 138 KV Line
Contingency: MIRA VISTA / YARNEN 345 KV
Loading: 107.8% (Rating: 707 MVA)
DFAX: 12.6%
Pre-project loading: 106.5%
12. Facility: PRINCE / HANFORD SUBSTATION 138 KV
Contingency: YARNEN SUBSTATION / MIRA VISTA 345 KV L
Loading: 107.3% (Rating: 261 MVA)
DFAX: 6.7%
Pre-project loading: 105.4%
13. Facility: BOBVILLE SUBSTATION / AVRA VALLEY 138 KV
Contingency: MCCARTNEY / YARNEN 345 KV
Loading: 106.3% (Rating: 445 MVA)
DFAX: 10.2%
Pre-project loading: 104.7%
14. Facility: CHUCKATUCK / KRAFT 138 KV
Contingency: MIRA VISTA / YARNEN 345 KV Line
Loading: 106.3% (Rating: 762 MVA)
DFAX: 12.6%
Pre-project loading: 105.1%
15. Facility: AVRA VALLEY / CHUCKATUCK 138 KV
Contingency: YARNEN / MIRA VISTA 345 KV
Loading: 105.7% (Rating: 792 MVA)
DFAX: 12.6%
Pre-project loading: 104.6%
16. Facility: SUFFOLK SUBSTATION / LTG MCCONAL 138 KV
Contingency: YARNEN / MCCARTNY 345 KV Line

Loading: 104.6% (Rating: 200 MVA)

DFAX: 29.9%

Pre-project loading: 93.9%

17. Facility: SANGER / COCHISE 138 KV

Contingency: MCCARTNEY / YARNEN 345 KV

Loading: 104.3% (Rating: 284 MVA)

DFAX: 20.8%

Pre-project loading: 99.0%

18. Facility: TOANO 345 kV/1 kV Transformer

Contingency: MIRA VISTA / YARNEN 345 KV

Loading: 103.1% (Rating: 704 MVA)

DFAX: 7.9%

Pre-project loading: 102.3%

19. Facility: SORENSEN / ALPINE 138 KV

Contingency: BOBVILLE / AVRA VALLEY 138 KV L

Loading: 102.0% (Rating: 237 MVA)

DFAX: 12.9%

Pre-project loading: 98.0%

20. Facility: SANGER / E LONGMEADOW 138 KV

Contingency: MCCARTNEY / YARNEN 345 KV

Loading: 101.9% (Rating: 566 MVA)

DFAX: 49.3%

Pre-project loading: 95.6%

21. Facility: LAKE KINGMAN / TILLMAN 138 KV

Contingency: YARNEN / MIRA VISTA SUBSTATION 345 KV

Loading: 100.3% (Rating: 301 MVA)

DFAX: 5.9%

Pre-project loading: 98.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CRITTENDEN / SMITHFIELD 138 KV	Rebuild line with higher-capacity conductor	\$16,875,000	4% (\$695,000)
TOANO SUBSTATION 138 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$6,975,000	1% (\$52,000)
MEYER RD / SORENSEN 138 KV L	Rebuild line with higher-capacity conductor	\$2,622,000	5% (\$126,000)
WERNER W / YARNEN SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$4,989,000	4% (\$225,000)
SMITHFIELD / HANFOD 138 KV L	Rebuild line with higher-capacity conductor	\$2,408,000	4% (\$99,000)
SUFFOLK / MYRTLE 138 KV line	Rebuild line with higher-capacity conductor	\$3,345,000	33% (\$1,115,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MYRTLE / SANGER 138 KV	Rebuild line with higher-capacity conductor	\$19,260,000	54% (\$10,395,000)
YARNEN 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$25,484,000	4% (\$1,147,000)
BOBVILLE / FOOTHILLS 138 KV line	Reconductor line	\$5,224,000	16% (\$838,000)
CARROLL / CRITTENDEN SUBSTATION 138 KV Line	Reconductor line	\$2,736,000	2% (\$55,000)
CRITTENDEN / KRAFT 138 KV	Reconductor line	\$2,581,000	2% (\$58,000)
HANFORD / PRINCE 138 KV	Reconductor line	\$837,000	4% (\$34,000)
AVRA VALEY / BOBVILLE 138 KV Line	Reconductor line	\$5,027,000	3% (\$133,000)
KRAFT SUBSTATION / CHUCKATUCK 138 KV	Reconductor line	\$3,087,000	2% (\$69,000)
AVRA VLY / CHUCKATUCK 138 KV L	Reconductor line	\$1,982,000	2% (\$44,000)
LTG MCCONAL / SUFFOLK 138 KV L	Reconductor line	\$1,481,000	33% (\$494,000)
SANGER SUBSTATION / COCHISE 138 KV L	Reconductor line	\$1,099,000	7% (\$78,000)
TOANO 345 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$13,460,000	1% (\$100,000)
SORENSEN / ALPINE 138 KV	Reconductor line	\$4,597,000	5% (\$221,000)
SANGER / E LONGMEADOW 138 KV	Reconductor line	\$2,685,000	10% (\$263,000)
TILLMAN SUBSTATION / LAKE KINGMAN 138 KV	Reconductor line	\$379,000	2% (\$8,000)
	Total allocated to project		\$16,249,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.